Band-1 Hadamard Order 2564 — Miyamoto (Kronecker H_640) Bonus; Gap 2052 Still OPEN

H_20/H_32/H_640/C_642/H_2564 PASS; 2564 Miyamoto-only bonus (not in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2644_Miyamoto`; `2026-09-15_Band1_Hadamard_2468_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_2564_miyamoto/`.
Cutoff. 2026-09-15 (~01:55Z).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 20, 32, 640 and 2564 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed H_{640}. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2564 is a **Miyamoto-only bonus** (not in SPK): Paley-I **FAIL** ($q = 2563 = 11 \cdot 233$ composite); Paley-II **FAIL** ($q = 1281$ composite); Kronecker **FAIL** ($2564 = 4 \cdot 641$, 641 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2644 on box. This increment constructs the optional Miyamoto-eligible target **order 2564** ($2564 = 4 \cdot 641$, 641 prime $\equiv 1 \pmod 4$). Seed H_{640} via Kronecker $H_{32} \otimes H_{20}$ (Sylvester × Paley type I $q = 19$). Symmetric Paley conference C_{642} over prime 641. $H_{2564} = \text{Miyamoto}(H_{640}, C_{642})$. Unit tests **24/24 PASS**. SPK next gap remains **2052** (OPEN). Order 2564 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2564; Miyamoto; Kronecker; Paley conference; constructive gap 2052; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{20} constructed and verified (Paley type I, $q = 19$). 2. H_{32} constructed and verified (Sylvester). 3. H_{640} constructed and verified (Kronecker $H_{32} \otimes H_{20}$). Paley-I/II for 640 **rejected**. 4. C_{642} constructed and verified (Paley conference, $q = 641$): $CC^\top = 641I$, zero diagonal. 5. H_{2564} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 6. SPK: $2564 \notin$ SPK (Miyamoto-only bonus). Paley-I, Paley-II, and Kronecker paths for 2564 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2564 = 4 \cdot 641$
$q = 641$	prime, $641 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{640}
Needed conference	order $642 = q + 1$
H_{640} path	Kronecker(H_{32} Sylvester, H_{20} Paley-I $q = 19$)
Paley-I for 640	**REJECTED**: $q' = 639$ not prime
Paley-II for 640	**REJECTED**: $q'' = 319$ not prime
Conference path	Symmetric Paley over prime 641
Paley-I for 2564	**REJECTED**: $q' = 2563 = 11 \cdot 233$ not prime
Paley-II for 2564	**REJECTED**: $q'' = 1281 = 3 \cdot 7 \cdot 61$ not prime
Kronecker for 2564	**REJECTED**: only 4×641; $641 \not\equiv 0 \pmod 4$

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{20}	Paley type I, $q = 19$	0	0.0002	**PASS**
H_{32}	Sylvester	0	0.0002	**PASS**
H_{640}	Kronecker(H_{32}, H_{20})	0	0.1292	**PASS**
C_{642}	Paley conference, $q = 641$	0	0.2951	**PASS**
H_{2564}	Miyamoto(H_{640}, C_{642})	0	7.2859	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh) ≈ 8.0617 s. Compressed ± 1 matrices saved as `H20_pm1.npz`, `H32_pm1.npz`, `H640_pm1.npz`, `C642_pm1.npz`, `H2564_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\{24, 284, 308, 508, 528, 540, 556, 568, 576, 600, 616, 660, 2004, 2036,$ $\cup$ verified $H_{640} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2564 does **not** enter SPK (classical paths fail).

Quantity	Value
Known count	880
2564 in SPK	**False** (Miyamoto-only bonus)
640 in SPK	True (admissible ≤ 2000; Kronecker-constructed on box)
2644 in SPK	True (prior; Paley II $q = 1321$)
2468 in SPK	True (prior; Paley I $q = 2467$)
2276 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (24/24)

Test	Result
`H20_paley_I_control` / `H4_sylvester` / `H32_sylvester`	PASS
`paley_II_q5_H12` / `paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6` / `paley_I_q11_H12`	PASS
`kronecker_H32xH20_H640`	PASS
`q641` / `n2564` / `H640` / `C642` formulae	PASS
`paley_I/II_H640_path_rejected`	PASS
`paley_I/II/kronecker_H2564_path_rejected`	PASS
`reject_all_ones_H8` / `conference_6_paley_control`	PASS
`miyamoto_eligible_2564_arith` / `conference_642_shape_expected`	PASS
`gap_2052_not_filled_by_this_ship`	PASS
`2564_miyamoto_only_bonus_arith`	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 641$, $q \equiv 1 \pmod 4$ (sample; $q = 661 \to 2644$ already shipped):

q	$n = 4q$	Need H_{q-1}	On-box path notes
653	2612	652	no quick SPK seed noted
673	2692	672	Kronecker(4x168)
677	2708	676	PaleyII(q=337)
701	2804	700	PaleyII(q=349)
709	2836	708	PaleyII(q=353)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct H_{2692} / H_{2708}, **or** Track B / Object 1 await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Full Hadamard conjecture: **OPEN**
- Clay-closed / finished-ToE: **false / not claimed**
- Gap 2052 filled: **NOT claimed** (OPEN)

- GATE-04: **LOCKED**
- RH / P≠NP / YM / Frankl / NS-solved / AC-solved: **not claimed**
- 2564 in SPK: **false** (Miyamoto-only bonus)

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2564_miyamoto
python3 verify_hadamard_2564.py
# expects: Unit tests 24/24; H20/H32/H640/C642/H2564 PASS; SPK audit rejects classical paths
# artifacts: results.json, literature_ledger.json, SUMMARY.txt, *.npz
```

Lab path. `/workspace/experiments/band1/hadamard_2564_miyamoto/`
PDF shelf. `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2564_Miyamoto.pdf`